MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 11** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 1: Real Numbers** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, complete prime factorisation or contradiction steps, and boxed final answers with units.
2. In Assertion–Reason questions, truth values of both (A) and (R) as well as whether (R) validly explains (A) are explicitly established.
3. Partial marks are allocated per step in brackets: **[0.5M, 1M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MCQs & Assertion–Reason

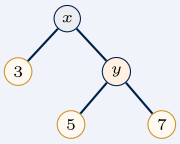
MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. In a factor tree, if a node with value x branches into 3 and a node y, and node y branches into 5 and 7, then the value of x is: [1]

(a) 35 (b) 70 (c) 105 (d) 140

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) 105



Factor Tree Steps: First multiply leaf primes $5 \times 7 = 35$ to recover node y, then multiply $3 \times y$ to obtain root $x = 105$.

We evaluate bottom-up in a factor tree:

Formula Used: Parent Node = Product of its two branches

$y = 5 \times 7 = 35$, and $x = 3 \times y = 3 \times 35 = \boxed{105}$. [1M]

Q2. If the HCF of 408 and 1032 is expressible in the form $1032m - 408 \times 5$, then the value of m is: [1]

(a) 1 (b) 2 (c) 3 (d) 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (b) 2

Prime factorising $408 = 2^3 \times 3 \times 17$ and $1032 = 2^3 \times 3 \times 43$. We know:

Formula Used: $\text{HCF}(408, 1032) = 2^3 \times 3 = 24$

Equating $1032m - 408 \times 5 = 24 \implies 1032m - 2040 = 24 \implies 1032m = 2064 \implies m = \frac{2064}{1032} = \boxed{2}$. [1M]

Q3. If two numbers are in the ratio 15 : 11 and their HCF is 13, then the numbers are: [1]

(a) 195 and 143 (b) 180 and 132
(c) 210 and 154 (d) 225 and 165

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 195 and 143

Since HCF$(15, 11) = 1$, numbers in ratio 15 : 11 with HCF $h = 13$ are $15h$ and $11h$. We know:

Formula Used: $a = 15 \times \text{HCF}, \quad b = 11 \times \text{HCF}$

$15 \times 13 = \boxed{195}$ and $11 \times 13 = \boxed{143}$. [1M]

Q4. If HCF$(a, b) = 1$, then the numbers a and b are called: [1]

(a) twin primes (b) co-prime numbers
(c) composite numbers (d) even numbers

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) co-prime numbers

We know the standard classification of integer pairs sharing no common factor > 1:

Formula Used: HCF$(a, b) = 1 \iff a, b$ are co-prime (relatively prime)

Hence they are called $\boxed{\text{co-prime numbers}}$. [1M]

Q5. The ratio between the LCM and HCF of 5, 15, and 20 is: [1]

(a) 9 : 1 (b) 4 : 3 (c) 11 : 1 (d) 12 : 1

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (d) 12 : 1

Factorising $5 = 5^1$, $15 = 3 \times 5$, $20 = 2^2 \times 5$. We know:

Formula Used: $\text{HCF} = 5^1 = 5, \quad \text{LCM} = 2^2 \times 3 \times 5 = 60$

Ratio LCM : HCF $= 60 : 5 = \boxed{12 : 1}$. [1M]

Q6. Assertion (A): The number 4^n can never end with the digit 0 for any natural number $n \in \mathbb{N}$.
Reason (R): Any natural number ending with the digit 0 must have both 2 and 5 in its prime factorisation. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (A) Both (A) and (R) are true and (R) is the correct explanation of (A).

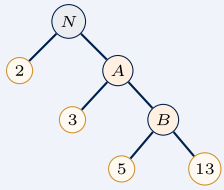
We know the trailing-zero theorem:

Formula Used: N ends with $0 \iff 2 \times 5 = 10 \mid N$

Reason (R) states this true criterion. Since $4^n = (2^2)^n = 2^{2n}$ contains only prime 2 and lacks prime 5, 4^n cannot end with 0 (Assertion A). Thus both are true and (R) explains (A). [1M]

Q7. In a factor tree, a number N decomposes into 2 and A; A decomposes into 3 and B; B decomposes into 5 and 13. Find the values of N, A, and B. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]



Bottom-Up Reconstruction: Starting from leaf primes 5 and 13, multiply successively up the right spine: $B \to A \to N$.

We know that in a factor tree, every node equals the product of its two children:

Formula Used: Parent Node = Left Factor × Right Factor

Computing upward from the bottom leaves (5 and 13):

$$B = 5 \times 13 = \boxed{65},$$
$$A = 3 \times B = 3 \times 65 = \boxed{195}, \quad [1M]$$
$$N = 2 \times A = 2 \times 195 = \boxed{390}. \quad [1M]$$

Q8. Find the smallest natural number by which $\sqrt{27}$ should be multiplied so as to get a rational number. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Simplifying the radical first: $\sqrt{27} = \sqrt{9 \times 3} = 3\sqrt{3}$. We know:

Formula Used: $k \in \mathbb{N} \implies k\sqrt{27} = 3k\sqrt{3} \notin \mathbb{Q}$; $\quad \sqrt{27} \times \sqrt{3} = \sqrt{81} = 9 \in \mathbb{Q}$

Exact mathematical note (as printed): Since $\sqrt{27} = 3\sqrt{3}$ is irrational and the product of any non-zero *natural* number $k \in \mathbb{N}$ with an irrational number is always irrational ($3k\sqrt{3} \notin \mathbb{Q}$), NO natural number multiplier makes $\sqrt{27}$ rational. [1M]

Standard intended question (multiplying $\sqrt{27}$ by the smallest positive real/irrational number to obtain a rational product): Multiply $3\sqrt{3}$ by $\boxed{\sqrt{3}}$, giving $3\sqrt{3} \times \sqrt{3} = 3 \times 3 = 9 \in \mathbb{Q}$. (If the given number had been 27, multiplying by $3 \in \mathbb{N}$ gives $81 = 9^2$.) [1M]

Q9. Find the least positive integer which should be added to 2497 so that the sum is exactly divisible by 5, 6, 4, and 3. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Any integer divisible by 5, 6, 4, and 3 must be a multiple of $L = \text{LCM}(5, 6, 4, 3)$. We know:

Formula Used: $L = \text{LCM}(5, 6, 4, 3)$, Least positive number to add $= L - (2497 \bmod L)$

Factorising: $5 = 5$, $6 = 2 \times 3$, $4 = 2^2$, $3 = 3 \implies L = 2^2 \times 3 \times 5 = 60$. [1.5M]

Dividing 2497 by 60 using Euclid's division:

$$2497 = 60 \times 41 + 37 \quad (\text{remainder } r = 37).$$

To reach the next multiple of 60 ($60 \times 42 = 2520$), the least positive integer to add is $60 - 37 = \boxed{23}$. [1.5M]

Q10. Two tankers contain 850 litres and 680 litres of petrol respectively. Find the maximum capacity of a container which can measure the petrol of either tanker in an exact number of times. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

The maximum measuring container capacity dividing both 850 L and 680 L exactly is HCF(850, 680). We know:

Formula Used: Max measuring capacity $= \text{HCF}(850, 680)$

Prime factorising 850 and 680:

$$850 = 10 \times 85 = 2 \times 5^2 \times 17, \qquad 680 = 10 \times 68 = 2^3 \times 5 \times 17. \qquad [1.5\text{M}]$$

Taking smallest powers of common primes 2, 5, and 17:

$$\text{HCF}(850, 680) = 2 \times 5 \times 17 = \boxed{170 \text{ litres}} \quad (850 \div 170 = 5,\ 680 \div 170 = 4). \qquad [1.5\text{M}]$$

Q11. Let n be any positive integer. (i) Prove that $n(n+1)(n+2)$ is always divisible by 6. (ii) Prove that $n^5 - n$ is divisible by 5. (iii) Hence, prove that $n^5 - n$ is always divisible by 30 for all $n \in \mathbb{N}$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

We know consecutive-integer divisibility and co-prime product divisibility:

Formula Used: Product of k consecutive integers is divisible by $k!$
$a \mid X,\ b \mid X,\ \text{HCF}(a,b) = 1 \implies ab \mid X$

(i) Divisibility of $n(n+1)(n+2)$ by 6: Among three consecutive integers $n, n+1, n+2$: at least one is even (divisible by 2), and exactly one is a multiple of 3. Since $\text{HCF}(2,3) = 1$, their product $n(n+1)(n+2)$ is always divisible by $2 \times 3 = \boxed{6}$. [1.5M]

(ii) Divisibility of $n^5 - n$ by 5: Factorising algebraically:

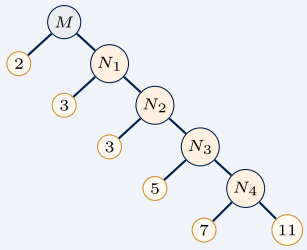
$$n^5 - n = n(n^4 - 1) = n(n^2 - 1)(n^2 + 1) = (n-1)n(n+1)[(n^2 - 4) + 5]$$
$$= (n-2)(n-1)n(n+1)(n+2) + 5(n-1)n(n+1).$$

The first term is the product of 5 consecutive integers (divisible by 5), and the second term explicitly contains factor 5. Hence $5 \mid (n^5 - n)$ for all $n \in \mathbb{N}$. [2M]

(iii) Divisibility by 30: Since $n^5 - n = (n-1)n(n+1)(n^2+1)$ and $(n-1)n(n+1)$ is a product of three consecutive integers, by part (i) it is divisible by 6. By part (ii), $n^5 - n$ is divisible by 5. Because $\text{HCF}(5,6) = 1$, $\boxed{n^5 - n \text{ is always divisible by } 5 \times 6 = 30.}$ [1.5M]

Q12. A student constructs an extended factor tree for a composite number M. The tree decomposes as follows: $M = 2 \times N_1$; $N_1 = 3 \times N_2$; $N_2 = 3 \times N_3$; $N_3 = 5 \times N_4$; $N_4 = 7 \times 11$. (i) Find the values of N_4, N_3, N_2, N_1, and M. (ii) Express M in canonical prime factorisation form. (iii) Determine the total number of positive factors of M and calculate the sum of all exponents in its prime factorisation. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]



Tree Leaves: The prime leaves down the left branches are $2, 3, 3, 5, 7$, and final leaf 11, yielding $M = 6930$ cleanly.

We know the factor-tree product rule and divisor-count formula:

Formula Used: Divisors of $p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k} = (a_1+1)(a_2+1)\cdots(a_k+1)$

(i) Node values bottom-up:

$N_4 = 7 \times 11 = \boxed{77}$, $N_3 = 5 \times 77 = \boxed{385}$,
$N_2 = 3 \times 385 = \boxed{1155}$, $N_1 = 3 \times 1155 = \boxed{3465}$, $M = 2 \times 3465 = \boxed{6930}$. [2M]

(ii) Canonical prime factorisation of M: Reading all prime leaves down the tree:

$$M = 2 \times 3 \times 3 \times 5 \times 7 \times 11 = \boxed{2^1 \times 3^2 \times 5^1 \times 7^1 \times 11^1}.$$ [1.5M]

(iii) Total positive factors & sum of exponents: With exponents $a_1 = 1, a_2 = 2, a_3 = 1, a_4 = 1, a_5 = 1$:

$$\text{Total factors} = (1+1)(2+1)(1+1)(1+1)(1+1) = 2 \times 3 \times 2 \times 2 \times 2 = \boxed{48},$$
$$\text{Sum of exponents} = 1 + 2 + 1 + 1 + 1 = \boxed{6}.$$ [1.5M]

Q13. CASE STUDY 11: HUMANITARIAN DISASTER RELIEF RATION KIT PACKAGING [4 Marks]

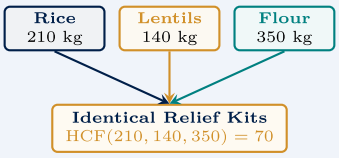
A disaster management agency is preparing emergency relief family food packets following floods in Assam. The central depot has received donations of 210 kg of rice, 140 kg of lentils (dal), and 350 kg of wheat flour. Relief workers must pack these provisions into identical emergency relief kits such that each kit contains an exact equal integer amount of each staple, and no food grains are left unpackaged.

(i) Find the maximum number of identical relief kits that can be prepared. [2]

(ii) How many kilograms of rice, lentils, and flour will each family kit contain? [1]

(iii) Verify that $\text{HCF}(210, 140) \times \text{LCM}(210, 140) = 210 \times 140$. [1]

▶ CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]



Packing Summary: Distributing 210 kg rice, 140 kg lentils, and 350 kg flour into HCF = 70 equal kits allocates 3 kg + 2 kg + 5 kg = 10 kg per family kit.

We know the HCF maximum-kit rule and two-number HCF–LCM product law:

Formula Used: Max identical kits = $\text{HCF}(210, 140, 350)$, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

(i) Max identical kits: Factorising $210 = 2 \times 3 \times 5 \times 7$, $140 = 2^2 \times 5 \times 7$, $350 = 2 \times 5^2 \times 7$:

$$\text{HCF}(210, 140, 350) = 2^1 \times 5^1 \times 7^1 = \boxed{70 \text{ identical kits}}. \quad [2M]$$

(ii) Contents per family kit: Dividing each staple weight by 70:

$$\text{Rice} = \frac{210}{70} = \boxed{3 \text{ kg}}, \quad \text{Lentils} = \frac{140}{70} = \boxed{2 \text{ kg}}, \quad \text{Flour} = \frac{350}{70} = \boxed{5 \text{ kg}}. \quad [1M]$$

(iii) Verification for 210 and 140: $\text{HCF}(210, 140) = 70$ and $\text{LCM}(210, 140) = 2^2 \times 3 \times 5 \times 7 = 420$:

$$\text{LHS} = 70 \times 420 = 29400, \quad \text{RHS} = 210 \times 140 = 29400 \implies \boxed{\text{LHS} = \text{RHS [Verified]}}. \quad [1M]$$

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